

DOROZHNY, Russian FederationWMO: **248260**Lat: **61.446N**Lon: **114.361E**Elev: **1160**StdP: **14.09**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.7	-42.5	-50.1	0.3	-45.3	-46.6	0.4	-42.0	12.2	0.8	11.5	-1.7	3.4	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	84.4	63.8	80.7	62.4	77.0	60.7	66.2	79.5	64.6	77.2	62.7	73.9	5.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	85.6	70.2	59.8	80.0	68.9	57.9	74.7	67.4	31.6	79.6	30.3	77.6	28.9	74.0	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.9	9.6	-49.3	90.1	5.6	3.1	-53.4	92.3	-56.6	94.1	-59.8	95.9	-63.8	98.1		
(5)			-48.5	69.1	4.5	2.1	-51.7	70.6	-54.3	71.9	-56.9	73.1	-60.1	74.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	-18.4	-10.8	10.3	27.3	42.9	59.6	62.1	58.0	41.8	22.7	-4.4	-15.5
	DBStd	31.24	15.38	15.24	13.92	9.78	8.40	7.93	6.91	6.58	7.75	10.00	14.44	15.99
	HDD50	10787	2121	1702	1231	680	254	13	3	14	262	845	1632	2030
	HDD65	15369	2586	2122	1696	1130	687	197	139	230	695	1310	2082	2495
	CDD50	991	0	0	0	0	33	301	378	262	17	0	0	0
	CDD65	97	0	0	0	0	1	34	49	13	0	0	0	0
	CDH74	1473	0	0	0	0	16	545	669	237	6	0	0	0
	CDH80	387	0	0	0	0	2	146	196	42	1	0	0	0
Wind	WSAvg	4.7	4.5	4.7	5.3	5.5	5.7	4.5	4.2	4.1	4.4	4.8	4.5	4.6
Precipitation	PrecAvg	16.8	0.8	0.6	0.7	0.8	1.4	2.0	2.5	2.3	1.8	1.6	1.3	0.9
	PrecMax	22.4	1.7	1.2	1.2	2.0	2.9	3.7	5.1	4.2	3.4	2.8	2.5	1.8
	PrecMin	13.3	0.4	0.3	0.2	0.3	0.9	0.9	0.7	1.1	0.6	0.6	0.4	0.3
	PrecStd	2.0	0.3	0.2	0.2	0.4	0.5	0.6	1.1	0.8	0.7	0.6	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	24.3	45.3	53.4	75.7	88.8	90.2	84.5	71.3	47.3	35.5	26.6
		MCWB	14.6	20.8	35.6	40.4	54.8	64.8	66.9	64.5	56.1	39.9	34.0	25.8
	2%	DB	10.8	19.0	39.4	49.4	69.5	84.1	85.2	80.0	63.5	43.0	28.0	19.6
		MCWB	9.7	17.0	32.2	38.6	51.4	62.2	65.8	62.9	50.8	38.1	26.9	18.5
	5%	DB	7.5	14.6	34.0	46.0	64.6	80.2	81.2	76.4	59.3	39.4	22.9	14.7
		MCWB	6.7	12.9	28.5	36.1	48.7	60.8	63.5	60.7	48.9	35.1	21.3	13.8
	10%	DB	3.5	10.5	30.0	42.6	59.3	76.2	77.1	72.5	54.7	36.2	18.1	9.6
		MCWB	2.8	9.1	26.0	34.4	45.5	59.1	61.9	59.8	46.7	33.0	17.2	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	21.3	36.5	41.7	56.3	66.7	69.0	66.6	56.6	41.7	34.3	25.7
		MCDB	15.3	23.7	44.3	50.4	73.6	83.5	84.0	78.4	66.9	46.3	35.1	26.5
	2%	WB	9.8	17.1	32.5	39.3	52.1	64.5	67.0	64.7	53.4	38.8	26.9	18.9
		MCDB	10.7	18.9	38.7	48.2	66.9	80.5	81.0	76.1	60.4	42.0	28.1	19.5
	5%	WB	6.7	13.0	29.0	36.9	49.4	62.7	65.5	62.8	50.1	35.5	21.4	14.0
		MCDB	7.6	14.6	33.4	44.6	62.6	77.1	78.1	72.9	56.4	38.4	22.7	14.8
	10%	WB	2.9	9.2	26.1	34.7	46.4	60.8	63.8	60.8	47.9	32.9	17.2	8.8
		MCDB	3.6	10.6	30.0	41.9	58.7	72.5	74.9	70.1	54.4	36.0	18.2	9.6
Mean Daily Temperature Range	5% DB	MDBR	12.6	16.1	19.9	19.0	18.9	22.0	20.9	21.6	16.4	11.5	13.4	12.8
		MCDBR	12.6	15.4	19.0	21.4	28.0	28.7	27.9	30.1	25.2	14.1	15.8	15.3
		MCWBR	12.1	14.0	13.7	13.9	14.7	12.1	11.1	13.8	13.9	10.2	15.8	15.2
	5% WB	MCDBR	12.2	15.4	17.3	18.6	25.6	24.8	24.5	24.1	21.7	12.9	15.2	15.6
		MCWBR	11.8	14.1	12.8	12.8	14.3	11.3	11.1	12.3	12.9	10.2	15.3	15.5
Clear-Sky Solar Irradiance	taub	0.231	0.268	0.296	0.330	0.358	0.398	0.437	0.411	0.324	0.283	0.228	0.195	
	taud	1.951	2.047	2.082	2.129	2.321	2.322	2.203	2.238	2.471	2.334	1.990	1.875	
	Ebn at Noon	144	218	256	271	273	263	246	242	252	219	143	101	
	Edh at Noon	13	24	33	39	35	36	40	36	23	18	12	8	
All-Sky Solar Radiation	RadAvg	123	423	960	1548	1652	1826	1631	1270	782	358	158	63	
	RadStd	6	25	48	91	114	135	129	77	76	38	17	3	